1 Algebraic Gauss tails and centered harmonic reflection

Define Bernoulli-coefficient polynomials by

$$D_j(a,b,s;c) = \frac{(-1)^{j+1}}{j(j+1)}\big(B_{j+1}(a-c)+B_{j+1}(b-c)\big) - \frac{(-1)^{j+1}}{j(j+1)}\big(B_{j+1}(a+b+s-c)+B_{j+1}(1-c)\big),$$

$$\sum_{m\geq 0} A_m(a,b,s;c)t^m = \exp\left(\sum_{j\geq 1} D_j(a,b,s;c)t^j\right).$$

For every fixed coefficient these are finite polynomial definitions.

Theorem 1.1 (Finite residue proof and full resonant tail)**.** *For all integers $r,k\geq 0$,*

$$A_{r+k}(a,b,-r;c) = \frac{(-1)^r}{r!}(1-a)_r(1-b)_r Q_{r,k}(a,b;c),$$

where the finite polynomial is

$$Q_{r,k}(a,b;c) = \sum_{\ell=0}^{k}\binom{k}{\ell}c^{k-\ell}\sum_{j=0}^{\ell}(-1)^j\left\{\begin{matrix}\ell\\ j\end{matrix}\right\}\frac{(a)_j(b)_j}{(r+1)_j}.$$

Its exponential generating function is $\exp(ct)\,{}_2F_1(a,b;r+1;1-\exp t)$ *near zero, and* $\partial_c Q_{r,k} = kQ_{r,k-1}$.

Proof. At $s=-r$ the total degree of A_r in a,b is at most $2r$. For $b=B\in\{1,\dots,r\}$, Bernoulli finite differences give

$$\sum_{m\geq 0} A_m(a,B,-r;c)t^m = \prod_{h=1}^{r-B}(1+(a-c-h)t)\prod_{h=1}^{B-1}(1+(h-c)t).$$

This polynomial has degree at most $r-1$. Symmetry supplies the roots $a=1,\dots,r$ also; hence $(1-a)_r(1-b)_r$ divides A_r. The leading degree $2r$ comes only from $D_1^r/r!$, whose leading part is $(-ab)^r/r!$. Thus $A_r=(-1)^r(1-a)_r(1-b)_r/r!$. This argument for the residue is purely finite algebra; $r=0$ is immediate.

For the tail, take $c=0$ and the formal Gamma-ratio Laurent expansion $\mathcal{W}(x)=x^{r-1}\sum_j A_j x^{-j}$. The Gamma recurrence gives

$$(x+1)(x+a+b-r)\mathcal{W}(x+1) = (x+a)(x+b)\mathcal{W}(x).$$

The coefficients in this formal expansion are exactly those defined above, by the finite log-Gamma expansion in Bernoulli polynomials. Remove its polynomial part and send x^{-k-1} to $t^k/k!$. If $T(t)=\sum_{k\geq 0}A_{r+k}t^k/k!$, the resulting equation is

$$(1-e^t)T'' + [a+b-r-1-(a+b)e^t]T' - abe^tT = 0.$$

Multiplication by x becomes differentiation, while the unit shift becomes multiplication by e^{-t}; discarded polynomial parts remain polynomial, so no tail terms are lost. Substituting $z=1-e^t$ gives Gauss's equation with parameters $(a,b;r+1)$. Its unique formal solution with constant coefficient A_r is $A_r\,{}_2F_1(a,b;r+1;z)$. Coefficient uniqueness divides only by $(j+1)(r+1+j)$, which never vanishes.

Finally, Bernoulli differentiation gives $\partial_c A_m = (s+m)A_{m-1}$. On the tail this transports its exponential generating function by e^{ct}. Expanding $(e^t-1)^j = j!\sum_{\ell\ge j}\left\{{\ell \atop j}\right\} t^\ell/\ell!$ yields the stated finite formula. □

Now put

$$Z_r(s,a) = \sum_{n\ge 0} \frac{[u^r]\prod_{j=1}^n (1+u/j)}{(n+a)^s}, \qquad \mathcal{B}(s,u;a) = \sum_{n\ge 0} \frac{(1+u)_n}{n!(n+a)^s},$$

initially for $\Re a > 0$, $\Re s > 1$ and u sufficiently small. The equality $Z_r = [u^r]\mathcal{B}$ holds in a common convergence domain.

Theorem 1.2 (Moving reflection and common centered zeros)**.** *For $0 < \Re a < 1$, the plus family $\mathcal{B}(s,u;a) + \mathcal{B}(s,u;1+u-a)$ is jointly holomorphic and zero at $(s,u) = (-2m,u)$ for $m \ge 0$ and u sufficiently small. The minus family has the corresponding property at $s = -2m-1$. In particular,*

$$C_r(s) = \sum_{k=0}^{r} \frac{(-1)^k (s)_k}{2^k k!} Z_{r-k}(s+k, 1/2)$$

is holomorphic and zero at $s = 0, -2, -4, \ldots$ for every $r \ge 0$.

Proof. The Mellin kernel is $e^{-at}(1-e^{-t})^{-1-u}$. Write its normalized heat factor as

$$e^{-at}\left(\frac{t}{1-e^{-t}}\right)^{1+u} = \sum_{j\ge 0} c_j(u,a) t^j.$$

Its reflection is $c_j(u, 1+u-a) = (-1)^j c_j(u,a)$. Uniform subtraction of K heat terms gives moving denominators $s-1-u+j$, with a jointly holomorphic remainder in $\Re(s-u) > 1-K$. The plus combination removes odd j, so it removes the only possible pole near $s = -2m, u = 0$. The minus combination removes the corresponding even j at $s = -2m-1$. Multiplication by the vanishing reciprocal Gamma proves the joint holomorphy and zeros. The uniform remainder follows by integrating an $O(t^K)$ Taylor remainder near zero and an exponentially decaying kernel at infinity.

The centered shift $(1+u)/2$ has even normalized heat factor $(t/(2\sinh(t/2)))^{1+u}$ and therefore the same even-integer zeros. Taylor expansion of the moving shift gives the displayed finite formula; joint holomorphy allows extraction of every coefficient at its zero. □

Corollary 1.3 (Convergent harmonic evaluation at every depth)**.** *For $|u| < 1$,*

$$\sum_{r\ge 0} C_r(2) u^r = \frac{\pi \tan(\pi u/2)}{u} \frac{\Gamma((1+u)/2)\Gamma(1-u)}{\Gamma((1-u)/2)}.$$

Its removable value at zero is $\pi^2/2$. Equivalently, $C_r(2) = \pi^2 h_r/2$, where $h_0 = 1$ and

$$r h_r = \sum_{j=1}^{r} \kappa_j h_{r-j}, \quad \kappa_1 = -2\log 2, \quad \kappa_j = \begin{cases} (3-2^{2-j})\zeta(j), & j \ge 2 \text{ even}, \\ (2^{1-j}-1)\zeta(j), & j \ge 3 \text{ odd}. \end{cases}$$

Proof. For $\Re u < 0$, $\mathcal{B}(1,u;a) = B(a,-u)$, and differentiation gives $\mathcal{B}(2,u;a) = -B(a,-u)[\psi(a) - \psi(a-u)]$. At $a = (1+u)/2$, digamma reflection reduces the bracket to $\pi\tan(\pi u/2)$. Normal convergence extends the identity to $|u| < 1$. Log-Gamma expansion at 1 and $1/2$, together with the Taylor expansion of $\log(\tan(\pi u/2)/(\pi u/2))$, gives the stated cumulants. □

The fixed-shift negative Laurent theorem and the analytic Gauss residue identity remain credited to their existing sources. The new algebraic proof answers the separate finite-proof question. At singular points, all reflection identities refer to the continued complete combination, not separately evaluated divergent summands. Neither S6 nor S8 is resolved by this insertion.